PAW

LoRa Configuration

868 MHz P2P Radio Parameters

Rationale, Airtime Calculation & ETSI EU868 Duty-Cycle Compliance

Version 1.0
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Target Hardware: RAKwireless RAK3172 / STM32WLE5

1. Overview

This document specifies the LoRa Point-to-Point (P2P) radio configuration used by the PAW collar for transmitting location packets to a gateway over a private 868 MHz link. It states the exact parameters configured in firmware, explains the rationale behind each value, calculates on-air time for the fixed 18-byte payload, and checks the configuration against ETSI EU868 duty-cycle and power limits. Section 6 flags one parameter — channel bandwidth — that is not explicitly set in the current firmware and should be confirmed before the airtime and duty-cycle figures here are treated as final.

2. Radio Parameters

Parameter	Value	Firmware Constant
Frequency	868 MHz (EU868)	`LORA_FREQUENCY = 868000000UL`
Radio Mode	LoRa Point-to-Point (P2P)	`api.lora.nwm.set()`
Spreading Factor	SF9	`LORA_SPREADING_FACTOR = 9`
Coding Rate	4/5	`LORA_CODING_RATE = 0`
Preamble Length	8 symbols	`LORA_PREAMBLE = 8`
TX Power	14 dBm	`LORA_TX_POWER = 14`
Bandwidth	Not explicitly set — RUI3 default assumed (125 kHz)	`— (see Section 6)`
Payload Size	18 bytes, fixed	`PACKET_LEN = 18`

All values other than bandwidth are set explicitly in hal_lora_init() via the RUI3 api.lora. interface and are reproduced verbatim from collar_firmware.cpp.*

3. Parameter Rationale

3.1 Frequency — 868 MHz

868 MHz sits in the EU863-870 MHz ISM band, the standard license-free allocation for LoRa devices operating in Europe. It requires no cellular subscription or carrier agreement, consistent with the collar's design goal of avoiding SIM/cellular connectivity entirely.

3.2 Spreading Factor — SF9

SF9 is a mid-range spreading factor: higher SFs (SF10–SF12) extend range and receiver sensitivity at the cost of significantly longer airtime and higher energy per packet, while lower SFs (SF7–SF8) shorten airtime but reduce link margin. SF9 was chosen as a balance point for a wearable tracker that needs reasonable range without materially increasing the active-radio time already budgeted in the power design. This is a tunable parameter — see Section 7.

3.3 Coding Rate — 4/5

4/5 is the least redundant of LoRa's forward-error-correction coding rates (4/5 through 4/8). It adds the minimum overhead needed for basic error correction, keeping airtime as short as possible; higher coding rates (4/6–4/8) trade additional airtime for more robustness against interference, which is not currently judged necessary for a short, private P2P link.

3.4 Preamble — 8 Symbols

8 symbols is the LoRa standard default preamble length. It is long enough for the receiver to reliably detect and synchronize to the incoming signal without adding unnecessary fixed overhead to every transmission.

3.5 TX Power — 14 dBm

14 dBm (25 mW) is a deliberate mid-range setting, not the radio's maximum. It also happens to align exactly with the ERP limit ETSI imposes on the primary EU868 sub-band (Section 5), so it is already compliant with the regulatory ceiling for that band rather than being throttled down from a higher default.

> **Implementation note:** This value should be revisited once real-world link-margin needs from field testing are known — headroom exists only in the g3 sub-band (10% duty cycle / 27 dBm), not in the band currently configured.

4. Payload & Airtime

The payload is fixed at 18 bytes (PACKET_LEN) regardless of event type — see the Binary Packet Specification document for the exact field layout. Because LoRa airtime scales directly with payload size for a given SF/CR/BW combination, keeping the packet small and fixed-length is itself a power- and compliance-relevant design choice: every transmission costs the same, predictable amount of on-air time.

4.1 Worked Airtime Calculation

Using the standard LoRa airtime formulas (symbol duration = 2^SF / BW; preamble time = (n_preamble + 4.25) × symbol duration; payload symbol count per the Semtech formula for SF9, CR 4/5, an 18-byte payload, and CRC enabled):

Assumption	Explicit header + CRC	Implicit header + CRC
Bandwidth = 125 kHz (RUI3 default)	≈ 185.3 ms	≈ 164.9 ms
Bandwidth = 250 kHz	≈ 92.7 ms	≈ 82.4 ms

> **Discrepancy noted:** Prior PAW documents (README.md, Power Optimization Strategy) cite an airtime of "~60–70 ms" for this payload. That figure is closest to the 250 kHz / implicit-header case above, not the 125 kHz explicit-header case that RUI3's P2P mode uses unless configured otherwise — see Section 6.

5. Regulatory Compliance — EU868 (ETSI EN 300 220)

The EU863-870 MHz ISM band is divided into sub-bands, each with its own duty-cycle and power ceiling under ETSI EN 300 220. The collar's configured frequency (868 MHz) and TX power (14 dBm) place it in sub-band g1:

Sub-band	Frequency Range	Max Duty Cycle	Max ERP
g1 (M)	868.0 – 868.6 MHz	1.0%	25 mW (14 dBm)
g2 (N)	868.7 – 869.2 MHz	0.1%	25 mW (14 dBm)
g3 (P)	869.4 – 869.65 MHz	10.0%	500 mW (27 dBm)
g4 (Q)	869.7 – 870.0 MHz	1.0%	25 mW (14 dBm)

Sub-band g1 permits 36 seconds of cumulative transmit time per hour (1% duty cycle) at up to 25 mW ERP (14 dBm) — exactly the TX power configured in firmware. At the ~185 ms airtime calculated in Section 4.1, the collar could transmit roughly 190 times per hour before approaching the duty-cycle ceiling; even at the ~70 ms figure used elsewhere, that rises to roughly 500 times per hour. The firmware's worst-case event rate from the Power Optimization Strategy document (348 events/day, ≈14.5/hour) sits well inside this margin under either airtime figure, so duty-cycle compliance is not at risk under the currently modeled usage — but the margin narrows considerably if SF or payload size increase in the future (Section 7).

6. Implementation Cross-Reference

Each configuration item above maps to a specific, verifiable construct in collar_firmware.cpp:

Configuration Item	Firmware Construct	Location
Frequency, SF, CR, preamble, TX power	`api.lora.pfreq / psf / pcr / ppl / ptp`	hal_lora_init()
P2P mode selection	`api.lora.nwm.set()`	hal_lora_init()
TX callback registration	`api.lora.registerPSendCallback()`	hal_lora_init()
Packet transmission	`api.lora.psend(len, packet, false)`	hal_lora_send()
Bench-test mock path	`HW_LORA_MOCK build switch`	hal_lora_send()

7. Open Items & Recommendations

- Confirm channel bandwidth: hal_lora_init() does not call an explicit bandwidth setter (e.g. api.lora.pbw). Confirm the RUI3 P2P default actually in effect on the RAK3172, and set it explicitly in firmware rather than relying on an unstated default — this single value changes the airtime estimate by roughly 2×.
- Reconcile the ~70 ms airtime figure used in README.md and the Power Optimization Strategy document against the worked calculation in Section 4.1 once bandwidth is confirmed, since it feeds directly into the battery-life tables in the Power Optimization Strategy document.
- Re-run the duty-cycle margin check in Section 5 against the confirmed airtime figure and against real field event rates once available, rather than the modeled worst case.
- Revisit SF9 / 14 dBm against measured field link margin once hardware testing begins; if range proves insufficient, sub-band g3 (869.4–869.65 MHz, 10% duty cycle, 27 dBm) offers substantially more headroom than g1, at the cost of moving off the currently configured frequency.
- If payload size or spreading factor change in a future protocol revision, re-run both the airtime calculation (Section 4.1) and the duty-cycle margin (Section 5) — neither is fixed once those inputs change.